HALOALKANES AND HALOARENES

NCERT Unit 6 | Class 12 Chemistry | Complete NEET Notes

1. INTRODUCTION & OVERVIEW

Haloalkane: Halogen on sp3 C (alkyl group) — R-X
Haloarene: Halogen on sp2 C of aromatic ring — Ar-X

C-X Bond Nature: Carbon is partially +ve (δ+), Halogen is partially -ve (δ−). Bond is POLAR.

⭐ NEET MCQ: Important Clinically Relevant Halogen Compounds
Chloramphenicol (contains Cl) — antibiotic for typhoid fever
Thyroxine (contains I) — body hormone; deficiency causes goiter
Chloroquine — treatment of malaria
Halothane — anaesthetic during surgery
Fully fluorinated compounds — potential blood substitutes in surgery
DDT — insecticide (first chlorinated organic insecticide)

2. CLASSIFICATION

2.1 Based on Number of Halogen Atoms

Type	Formula/Example	Haloalkane	Haloarene
Mono	1 halogen	CnH2n+1X	C6H5X
Di	2 halogens	CH2X-CH2X	o/m/p-C6H4X2
Poly (Tri, Tetra...)	3+ halogens	CHX3, CX4	C6H3X3

2.2 Based on sp3 C-X Bond

(a) Alkyl Halides (R-X) — Haloalkanes

General formula: **CnH2n+1X** — Homologous series

Type	Structure	Example	NEET Trick
Primary (1°)	RCH2-X (1 alkyl on C)	CH3CH2Cl	Most SN2 reactive
Secondary (2°)	R2CH-X (2 alkyls on C)	CH3CHClCH3	SN1 or SN2

Tertiary (3°)	R3C-X (3 alkyls on C)	(CH3)3CCl	Most SN1 reactive

(b) Allylic Halides

Definition: Halogen on sp3-C adjacent to C=C (allylic carbon)

Example: CH2=CH-CH2Cl (allyl chloride / 3-Chloroprop-1-ene)

- **High SN1 reactivity** — allylic carbocation stabilised by resonance

(c) Benzylic Halides

Definition: Halogen on sp3-C attached to aromatic ring

Example: C6H5-CH2Cl (benzyl chloride / Chloromethylbenzene)

- **High SN1 reactivity** — benzylic carbocation stabilised by resonance

2.3 Based on sp2 C-X Bond

(a) Vinylic Halides

Definition: Halogen directly on sp2-C of C=C bond

Example: CH2=CHCl (vinyl chloride / Chloroethene)

- **LEAST reactive** towards nucleophilic substitution (bond has partial double bond character)

(b) Aryl Halides (Haloarenes)

Definition: Halogen directly on sp2-C of aromatic ring

Example: C6H5Cl (chlorobenzene / Bromobenzene)

- **Very LESS reactive** towards nucleophilic substitution (4 reasons — see section 7)

⭐ REACTIVITY ORDER FOR NEET (Must Memorise)
SN2 Reactivity: CH3X > 1° > 2° > 3° (methyl most reactive, tertiary least)
SN1 Reactivity: 3° > 2° > 1° > CH3X (tertiary most reactive)
For SAME alkyl group: R-I > R-Br > R-Cl > R-F (both SN1 and SN2)
Allylic = Benzylic — both show HIGH SN1 reactivity (resonance stabilised carbocation)
Vinylic, Aryl Halides — LEAST reactive towards nucleophilic substitution

3. NOMENCLATURE (IUPAC)

3.1 Haloalkanes — Rules

1. Choose longest chain containing halogen as principal chain
2. Number from end giving halogen LOWEST locant
3. Halogen named as prefix: fluoro-, chloro-, bromo-, iodo-

4. If multiple halogens: alphabetical order for naming (bromo before chloro)

Common Name	IUPAC Name	Classification
n-Propyl bromide	1-Bromopropane	1° alkyl halide
Isopropyl chloride	2-Chloropropane	2° alkyl halide
tert-Butyl bromide	2-Bromo-2-methylpropane	3° alkyl halide
Vinyl chloride	Chloroethene	Vinylic halide
Allyl bromide	3-Bromopropene	Allylic halide
Benzyl chloride	Chlorophenylmethane	Benzylic (1°)
Methylene chloride	Dichloromethane	Polyhalogen
Chloroform	Trichloromethane	Polyhalogen
Carbon tetrachloride	Tetrachloromethane	Polyhalogen

3.2 Gem and Vic Dihalides

Type	Definition	Common Name	IUPAC Name	Example
gem-dihalide	Both X on SAME carbon	Alkylidene halide	1,1-dihaloalkane	CH3CHCl2 (Ethylidene chloride / 1,1-Dichloroethane)
vic-dihalide	X on ADJACENT carbons	Alkylene dihalide	1,2-dihaloalkane	CH2Cl-CH2Cl (Ethylene dichloride / 1,2-Dichloroethane)

3.3 Dihalobenzenes — Common vs IUPAC

Common System	IUPAC System
o- (ortho) = 1,2 positions	1,2-Dichlorobenzene
m- (meta) = 1,3 positions	1,3-Dichlorobenzene
p- (para) = 1,4 positions	1,4-Dichlorobenzene (highest m.p.)
⭐ **NEET Trick: p-isomer has HIGHEST melting point**	
p-Dichlorobenzene: highest m.p. (symmetrical molecule fits crystal lattice better)	
All three dihalobenzene isomers have nearly SAME boiling points	
para > ortho ~ meta in terms of melting point	

4. NATURE OF C-X BOND

Bond	Bond Length (pm)	Bond Enthalpy (kJ/mol)	Dipole Moment (Debye)	Key Trend

CH3-F	139 (shortest)	452 (strongest)	1.847	Strongest bond, hardest to break
CH3-Cl	178	351	1.860 (highest!)	Highest dipole moment
CH3-Br	193	293	1.830	Moderate
CH3-I	214 (longest)	234 (weakest)	1.636	Weakest bond, easiest SN1/SN2

⭐ **NEET MCQ Points on C-X Bond**
Bond Length: C-F < C-Cl < C-Br < C-I (fluorine smallest, iodine largest halogen)
Bond Enthalpy: C-F > C-Cl > C-Br > C-I (C-F strongest, C-I weakest)
Dipole Moment: C-Cl > C-F > C-Br > C-I (C-Cl has HIGHEST dipole moment!)
R-F most stable (hardest to break) but R-I most reactive in SN1/SN2
For halogen compounds: reactivity R-I > R-Br > R-Cl >> R-F

5. PREPARATION OF HALOALKANES

5.1 From Alcohols (Most Important Method)

General: R-OH + Halogenating agent → R-X + byproduct

From Alcohols + HCl	**Type: Nucleophilic Substitution**
R-OH + HCl --[ZnCl2 catalyst]--> R-Cl + H2O	
Conditions: ZnCl2 catalyst needed for primary & secondary; tertiary: just conc. HCl at RT (no catalyst needed)	
Product/Note: Order of reactivity: 3° > 2° > 1°	

From Alcohols + HBr	**Type: Nucleophilic Substitution**
R-OH + HBr(48%) --[heat]--> R-Br + H2O	
Conditions: Constant boiling HBr (48%)	
Product/Note: Lucas test uses ZnCl2 + conc. HCl	

From Alcohols + HI	**Type: Nucleophilic Substitution**
R-OH + NaI + H3PO4 (95%) --[heat]--> R-I	
Conditions: Heated with NaI in 95% orthophosphoric acid	
Product/Note: Note: H2SO4 NOT used with KI — it oxidises I- to I2	

From Alcohols + PCl3 / PCl5	**Type: Nucleophilic Substitution**
3R-OH + PCl3 → 3R-Cl + H3PO3 | R-OH + PCl5 → R-Cl + POCl3 + HCl	
Conditions: Phosphorus halides; PBr3 and PI3 made in-situ (red P + Br2 or I2)	

Product/Note: Phosphorus tribromide/triiodide generated in-situ

From Alcohols + SOCl2 (BEST METHOD)	Type: Nucleophilic Substitution
R-OH + SOCl2 → R-Cl + SO2 + HCl	
Conditions: Thionyl chloride — PREFERRED method	
Product/Note: PURE alkyl halide obtained because SO2 and HCl are gases and escape. No side products.	

⭐ WHY SOCl2 is the BEST method — NEET Favourite
SOCl2 = Thionyl chloride; reaction gives R-Cl + SO2(g) + HCl(g)
Both byproducts are GASES — escape from reaction mixture
Product is PURE (no separation needed)
Methods using PCl5 or HCl leave ionic byproducts that contaminate product
This is the standard answer for 'which is the best reagent to convert R-OH to R-Cl?'

5.2 From Hydrocarbons

Method 1: Free Radical Halogenation of Alkanes

Free Radical Halogenation	Type: Free Radical Substitution (Chain Reaction)
CH3CH2CH2CH3 + Cl2 --[UV light/heat]--> CH3CH2CH2CH2Cl + CH3CH2CHClCH3	
Conditions: Cl2, UV light or heat — dark gives no reaction	
Product/Note: Gives MIXTURE of mono and polyhaloalkanes — difficult to separate; yield of single compound is LOW	

⭐ Free Radical Halogenation — NEET Points
Condition: UV light or heat (NOT dark)
Gives complex mixture — difficult to separate pure compound
Reactivity of halogens: F2 > Cl2 > Br2 > I2
Selectivity: I2 > Br2 > Cl2 > F2 (inverse of reactivity)
Chlorination gives more isomers (less selective), bromination more selective

Method 2: From Alkenes — Addition of HX (Markovnikov's Rule)

Addition of HX to Alkene	Type: Electrophilic Addition
CH3-CH=CH2 + HI → CH3-CHI-CH3 (major, Markovnikov) + CH3-CH2-CH2I (minor)	
Conditions: No catalyst needed; HX adds across double bond	
Product/Note: Markovnikov: H goes to carbon with MORE H atoms; X goes to carbon with	

FEWER H atoms

Anti-Markovnikov Addition (Peroxide Effect)	Type: Free Radical Addition
CH3CH=CH2 + HBr --[Peroxide]--> CH3-CH2-CH2Br (1° product, anti-Markovnikov)	
Conditions: Peroxide / ROOR / hv — ONLY for HBr (NOT HCl or HI)	
Product/Note: Peroxide gives anti-Markovnikov product. Only HBr shows this — HCl and HI do NOT.	

Method 3: Addition of X2 to Alkenes (Vicinal Dihalides)

Addition of Halogen to Alkene	Type: Electrophilic Addition
CH2=CH2 + Br2 --[CCl4]--> BrCH2-CH2Br (vic-dibromide)	
Conditions: Bromine in CCl4 — reddish brown colour discharged	
Product/Note: Used as TEST for unsaturation — reddish brown colour of Br2 disappears; product vic-dibromide is colourless	

5.3 Halogen Exchange Reactions

Finkelstein Reaction	Type: Halogen Exchange (SN2)
R-Cl + NaI --[dry acetone]--> R-I + NaCl(precipitates)	
Conditions: NaI in dry acetone; NaCl precipitates out (insoluble in acetone)	
Product/Note: Used to prepare ALKYL IODIDES from alkyl chlorides/bromides; NaCl precipitates — Le Chatelier's principle drives forward	

Swarts Reaction	Type: Halogen Exchange
R-Cl + AgF (or Hg2F2, CoF2, SbF3) --[heat]--> R-F + AgCl	
Conditions: Metallic fluoride: AgF, Hg2F2, CoF2 or SbF3	
Product/Note: Used to prepare ALKYL FLUORIDES — cannot be made by direct electrophilic substitution on benzene	

⭐ NEET Memory Trick: Named Reactions Summary
Finkelstein = R-Cl/Br + NaI (dry acetone) → R-I (iodide exchange)
Swarts = R-Cl/Br + metallic fluoride → R-F (fluoride exchange)
Sandmeyer = ArN2+Cl- + CuCl/CuBr → Ar-Cl/Br (for aryl halides)
Balz-Schiemann = ArN2+BF4- → Ar-F + N2 + BF3 (aryl fluoride)
Wurtz = 2R-X + 2Na (dry ether) → R-R + 2NaX
Wurtz-Fittig = R-X + Ar-X + 2Na (dry ether) → R-Ar + 2NaX
Fittig = 2Ar-X + 2Na → Ar-Ar (diphenyl) + 2NaX

6. PREPARATION OF HALOARENES

6.1 Electrophilic Aromatic Substitution (Halogenation)

Halogenation of Benzene/Arene	Type: Electrophilic Aromatic Substitution (EAS)
C6H6 + Cl2 --[Fe or FeCl3, dark]--> C6H5Cl + HCl (major: chlorobenzene)	
Conditions: Lewis acid catalyst: Fe or FeCl3 (for Cl2/Br2); dark conditions	
Product/Note: Fluorine: too reactive — not prepared this way. Iodine: reaction reversible — needs oxidising agent (HNO3 or HIO4) to oxidise HI formed	

On Toluene: Gives o-halotoluene and p-halotoluene (ortho and para products, since -CH3 is o,p-director)

6.2 From Amines — Sandmeyer's Reaction

Sandmeyer's Reaction (Cl and Br)	Type: Nucleophilic Substitution via Diazonium Salt
ArNH2 + NaNO2 + HCl (273-278K) → [ArN2+Cl-] + CuCl → ArCl + N2	
Conditions: Cold (273-278K); CuCl for Cl, CuBr for Br	
Product/Note: Diazonium salt formed first; then Cu(I) halide replaces -N2+ with halide. Used for specific aryl halides.	

For Aryl Iodide	Type: Nucleophilic Substitution
ArN2+X- + KI → ArI + N2 + KX	
Conditions: No CuX needed for iodide — just KI	
Product/Note: Iodine is unique — does not require Cu catalyst	

Balz-Schiemann (For Aryl Fluoride)	Type: Thermal Decomposition
ArNH2 → ArN2+BF4- --[heat]--> ArF + N2 + BF3	
Conditions: HBF4 used; heated to decompose	
Product/Note: Only method for ArF — cannot use electrophilic substitution (F2 too reactive)	

⭐ Sandmeyer vs Direct Halogenation — NEET Points
Direct electrophilic halogenation: only for Cl and Br (F too reactive; I reversible)
Sandmeyer: Cl and Br (using CuCl or CuBr as catalyst)
Gattermann reaction: ArNH2 → ArX using Cu(powder) + HX (not cuprous halide)

Aryl iodides: Sandmeyer with KI (no Cu needed)
Aryl fluorides: Balz-Schiemann reaction (via diazonium fluoroborate)
Aryl halides CANNOT be made from ArOH (C-O bond has partial double bond character — too strong)

7. PHYSICAL PROPERTIES

7.1 Physical State & Colour

- Alkyl halides are COLOURLESS when pure
- Bromides and iodides develop colour on exposure to light
- Many volatile halogen compounds have SWEET smell
- Methyl chloride, methyl bromide, ethyl chloride — gases at room temperature
- Higher members — liquids or solids

7.2 Boiling Points

- **Key reason:** Polar molecules; stronger dipole-dipole & van der Waals forces vs parent hydrocarbons
- **For same alkyl group:** R-I > R-Br > R-Cl > R-F (increases with halogen size/mass)
- **Branching effect:** Boiling point DECREASES with branching (less surface area)
- *Example:* n-Butyl bromide (375K) > sec-Butyl bromide (364K) > tert-Butyl bromide (346K)
- **Dihalobenzene isomers:** Nearly same boiling points, BUT para-isomer has HIGHEST melting point (symmetry → better crystal packing)

7.3 Solubility

- Haloalkanes — VERY SLIGHTLY soluble in water
- Completely soluble in ORGANIC solvents

Reason for low water solubility: Energy required to break H-bonds in water > energy released from new haloalkane-water interactions. New interactions are weaker than original H-bonds.

7.4 Density

- Bromo, iodo and polychloro derivatives — HEAVIER than water (density > 1)
- Density increases with: more carbon atoms, more halogen atoms, heavier halogen

Compound	Density (g/mL)	Compound	Density (g/mL)
n-C3H7Cl	0.89 (lighter than water)	CH2Cl2	1.336
n-C3H7Br	1.335	CHCl3	1.489
n-C3H7I	1.747	CCl4	1.595

8. CHEMICAL REACTIONS OF HALOALKANES

Three main categories: (1) Nucleophilic Substitution (2) Elimination (3) Reaction with Metals

8.1 NUCLEOPHILIC SUBSTITUTION REACTIONS

General: Nu:- + R-X → R-Nu + X- (halogen = leaving group)

Reagent (Nu-)	Nucleophile	Product	Class of Product
NaOH/KOH (aq)	HO-	ROH	Alcohol
H2O	H2O	ROH	Alcohol
NaOR' (sodium alkoxide)	R'O-	ROR'	Ether
NaI	I-	R-I	Alkyl iodide (Finkelstein)
NH3	NH3	RNH2	Primary amine
R'NH2	R'NH2	RNHR'	Secondary amine
KCN	CN- (via C)	RCN	Nitrile (alkyl cyanide)
AgCN	Ag-CN: (via N)	RNC	Isonitrile (isocyanide)
KNO2	O=N-O- (via O)	R-O-N=O	Alkyl nitrite
AgNO2	Ag-O-N=O (via N)	R-NO2	Nitroalkane
R'COOAg	R'COO-	R'COOR	Ester
LiAlH4	H-	RH	Hydrocarbon (reduction)
Mg (dry ether)	Mg metal	RMgX	Grignard reagent

⭐ AMBIDENT NUCLEOPHILES — Very Important for NEET
Ambident = TWO sites of nucleophilic attack
CN-: attacks via C → gives nitrile (RCN); attacks via N → gives isonitrile (RNC)
KCN gives nitrile (cyanide) as MAJOR product — C-C bond more stable than C-N
AgCN gives isonitrile (isocyanide) as MAJOR product — AgCN is covalent; N is free to donate
NO2-: attacks via O → alkyl nitrite (R-O-N=O); via N → nitroalkane (R-NO2)
KNO2 gives alkyl nitrite; AgNO2 gives nitroalkane
Trick: KCN/KNO2 = ionic → thermodynamic product (cyanide/nitrite); AgCN/AgNO2 = covalent → kinetic product

8.1.1 SN2 — Substitution Nucleophilic Bimolecular

- **Rate:** Rate = k[R-X][Nu-] — depends on BOTH reactants (2nd order kinetics)
- **Mechanism:** Single step, no intermediate; CONCERTED process
- **Stereochemistry:** INVERSION OF CONFIGURATION (Walden Inversion) — like umbrella turning inside out

- **Transition state:** C bonded to 5 atoms simultaneously; pentacoordinate carbon
- **Reactivity order:** CH3X > 1° > 2° > 3° (methyl most reactive, tertiary unreactive)
- **Steric effect:** Bulky groups around C hinder nucleophile approach → decreases SN2 rate
- **Rate:** Methyl (30) : Ethyl (1) : Isopropyl (0.02) : tert-Butyl (≈0)

Proposed by: Edward Davies Hughes and Sir Christopher Ingold (1937)

8.1.2 SN1 — Substitution Nucleophilic Unimolecular

- **Rate:** Rate = k[R-X] — depends on substrate only (1st order kinetics)
- **Mechanism:** TWO steps — Step 1 (slow, rate-determining): R-X → R+ + X-; Step 2 (fast): R+ + Nu- → R-Nu
- **Intermediate:** CARBOCATION (sp2, planar)
- **Stereochemistry:** RACEMISATION — nucleophile can attack from either face of planar carbocation
- **Reactivity order:** 3° > 2° > 1° > CH3X (tertiary most reactive — most stable carbocation)
- **Solvent:** Polar protic solvents (water, alcohol, acetic acid) — stabilise ions through solvation
- **Step 1 is slow & reversible:** C-Br bond breaks; energy from solvation of halide ion by protic solvent

⭐ SN1 vs SN2 Comparison — NEET Essential
SN2: 2nd order kinetics | single step | inversion of config | favoured by 1° halides
SN1: 1st order kinetics | 2 steps | racemisation | favoured by 3° halides
SN2: bulky nucleophile = DECREASES rate; strong nucleophile = favours SN2
SN1: stable carbocation intermediate; 3° > allylic/benzylic > 2° > 1°
SN2 favoured: polar APROTIC solvents (DMSO, acetone, DMF)
SN1 favoured: polar PROTIC solvents (water, alcohol, acetic acid)
Primary halides: prefer SN2; Tertiary halides: prefer SN1
Secondary halides: SN1 or SN2 depending on nucleophile/conditions

8.1.3 Stereochemical Aspects — Optical Activity & Chirality

Key Definitions

- **Optically active compound:** Rotates plane of plane-polarised light
- **Dextrorotatory (+, d):** Rotates light CLOCKWISE (to right)
- **Laevorotatory (-, l):** Rotates light ANTICLOCKWISE (to left)
- **Polarimeter:** Instrument to measure angle of rotation
- **Chiral:** Non-superimposable on its mirror image (like left and right hands)
- **Achiral:** Superimposable on its mirror image
- **Asymmetric carbon (stereocentre):** C with 4 DIFFERENT groups attached
- **Enantiomers:** Stereoisomers that are non-superimposable mirror images of each other
- **Racemic mixture (dl or ±):** 50:50 mixture of enantiomers; OPTICALLY INACTIVE (rotations cancel)
- **Racemisation:** Conversion of optically active compound to racemic mixture

⭐ Chirality — NEET MCQ Points
Asymmetric carbon: 4 different groups; makes molecule chiral
Enantiomers: same physical properties (m.p., b.p., density) but OPPOSITE optical rotation
Racemic mixture: optically INACTIVE (net rotation = 0)
SN2 → Inversion of configuration (Walden Inversion)
SN1 → Racemisation (carbocation is achiral/planar, attack from both sides)
Sign of optical rotation NOT directly related to absolute configuration (R/S)
Examples of chiral molecules: 2-chlorobutane, 2-bromopropanoic acid, BrClCHI
Louis Pasteur (1848) first observed mirror image crystals → foundation of stereochemistry
Van't Hoff & Le Bel (1874) independently proposed tetrahedral arrangement of carbon

Inversion, Retention, Racemisation

Process	Mechanism	Configuration Change	Example
Inversion	SN2	Mirror image product obtained	(-)2-bromooctane + NaOH → (+)octan-2-ol
Retention	No bond to chiral C broken	Same configuration	(-)-2-methylbutan-1-ol + HCl → (+)-1-Chloro-2-methylbutane
Racemisation	SN1	50:50 mixture (d + l)	Optically active 2-bromobutane → (±)butan-2-ol

8.2 ELIMINATION REACTIONS (β-Elimination)

Dehydrohalogenation (β-Elimination)	Type: Beta-Elimination (E2 mechanism under these conditions)
R-CH2-CH2-X + KOH (alcoholic) --[heat]--> R-CH=CH2 + KX + H2O	
Conditions: Alcoholic KOH + heat; alpha carbon = C bearing X; beta carbon = adjacent C	
Product/Note: H removed from beta-carbon; X leaves from alpha-carbon; alkene formed	

Zaitsev's Rule (Saytzeff's Rule)

"In dehydrohalogenation reactions, the preferred product is the alkene with the GREATER number of alkyl groups attached to the doubly bonded carbon atoms." (more substituted alkene is major product)

Example: 2-Bromopentane + alc. KOH → Pent-2-ene (81%, major) + Pent-1-ene (19%, minor)

📌 Elimination vs Substitution — Competition
Haloalkane + base/nucleophile: SN1, SN2 OR Elimination compete
1° halide: favours SN2 (less steric) with strong small nucleophiles

3° halide: favours SN1 or Elimination (bulky nucleophile → acts as base)
2° halide: SN2 or Elimination depending on base strength
Bulky nucleophile (e.g., t-BuOK): favours ELIMINATION over substitution
Strong small nucleophile (e.g., NaOH, CN-): favours SUBSTITUTION
Aqueous KOH → Substitution (alcohol formed)
Alcoholic KOH → Elimination (alkene formed)

8.3 REACTIONS WITH METALS

Grignard Reagent (RMgX)

Formation of Grignard Reagent	Type: Organometallic Compound Formation
R-X + Mg --[dry ether]--> R-MgX (alkyl magnesium halide)	
Conditions: Mg metal in DRY ETHER; anhydrous conditions essential	
Product/Note: C-Mg bond: covalent but highly polar (δ- on C, δ+ on Mg); Mg-X bond: ionic	

- **Victor Grignard (1900):** discovered; Nobel Prize in Chemistry 1912
- **Why DRY conditions:** Grignard reagent reacts with ANY proton source (even water, alcohol, amine) to give hydrocarbon

RMgX + H2O → RH + Mg(OH)X (destroyed by moisture)

Wurtz Reaction

Wurtz Reaction	Type: Organometallic / Coupling Reaction
2R-X + 2Na --[dry ether]--> R-R + 2NaX	
Conditions: Na metal in dry ether	
Product/Note: Gives alkane with DOUBLE the carbon atoms; used to prepare higher alkanes from lower halides	

📌 Grignard & Wurtz — NEET Points
Grignard: RMgX — prepared in DRY ether to prevent decomposition by water
Grignard reacts with any proton source → hydrocarbon (RH)
Grignard C-Mg bond is polar covalent; very nucleophilic carbon
Wurtz: gives R-R (double carbons); not useful for odd-carbon alkanes
Wurtz fails with unsymmetrical mixtures (gives mixture of products)

9. REACTIONS OF HALOARENES

9.1 Nucleophilic Substitution — Why LESS Reactive?

Haloarenes: VERY LESS reactive towards nucleophilic substitution (compared to haloalkanes)

4 Reasons for Low Reactivity:

5. **Resonance Effect:** Lone pairs on halogen in conjugation with pi electrons of ring → C-X bond has PARTIAL DOUBLE BOND CHARACTER → harder to break
6. **Hybridisation:** C bearing X is sp2 in haloarene (more electronegative) vs sp3 in haloalkane — sp2 C holds electron pair more tightly → C-X bond shorter (169 pm vs 177 pm in haloalkane) → harder to break
7. **Phenyl cation instability:** If C-X broke, phenyl cation formed — NOT stabilised by resonance → SN1 impossible
8. **Electron-rich ring:** Repulsion between incoming electron-rich nucleophile and electron-rich benzene ring

📌 Aryl vs Alkyl Halide Reactivity — NEET Comparison
Aryl C-X: sp2, 169 pm, partial double bond — LESS reactive
Alkyl C-X: sp3, 177 pm, pure single bond — MORE reactive
Haloarene with -NO2 at ortho/para: MUCH more reactive (activates ring)
-NO2 at META: NO effect on reactivity
Most reactive aryl halide: 2,4,6-trinitro chlorobenzene (picyl chloride)

Replacement by -OH (Nucleophilic)

Chlorobenzene to Phenol	Type: Nucleophilic Aromatic Substitution
C6H5Cl + NaOH(aq) --[623K, 300 atm]--> C6H5OH + NaCl	
Conditions: Very harsh conditions: 623K, 300 atm pressure	
Product/Note: Vigorous conditions show low reactivity of aryl halides	

Effect of -NO2 Groups	Type: Nucleophilic Aromatic Substitution (Addition-Elimination)
o-/p-NO2-C6H4Cl + NaOH --[443K]--> o-/p-NO2-C6H4OH (easier!)	
Conditions: -NO2 at ortho/para; much milder conditions	
Product/Note: -NO2 withdraws electron density → ring is more electrophilic → nucleophile attack easier. Meta-NO2: no effect.	

9.2 Electrophilic Aromatic Substitution (EAS) Reactions

Halogen is o,p-director but ring DEACTIVATOR

Inductive effect (−I): Withdraws electrons from ring → deactivation (reactivity reduced vs benzene)
Resonance effect (+M): Donates electrons back via lone pairs → directs to o and p positions
Net result: Reactivity controlled by inductive effect; Orientation controlled by resonance

Reactions of Haloarenes:

Halogenation of Halobenzene	Type: Electrophilic Aromatic Substitution
C6H5Cl + Cl2 --[FeCl3]--> 1,4-dichlorobenzene (Major) + 1,2-dichlorobenzene (Minor)	
Conditions: FeCl3 catalyst (anhydrous Lewis acid)	
Product/Note: Para product is MAJOR (less steric hindrance than ortho)	

Nitration of Halobenzene	Type: Electrophilic Aromatic Substitution
C6H5Cl + HNO3 --[conc. H2SO4]--> 1-Chloro-4-nitrobenzene (Major) + 1-Chloro-2-nitrobenzene (Minor)	
Conditions: Conc. HNO3 + conc. H2SO4	
Product/Note: Para product (1-Chloro-4-nitrobenzene) is major product	

Sulphonation of Halobenzene	Type: Electrophilic Aromatic Substitution
C6H5Cl + H2SO4 --[heat]--> 4-Chlorobenzenesulphonic acid (Major) + 2-Chlorobenzenesulphonic acid (Minor)	
Conditions: Conc. H2SO4, heat	
Product/Note: Para product major	

Friedel-Crafts Alkylation	Type: Electrophilic Aromatic Substitution
C6H5Cl + CH3Cl --[AlCl3]--> 1-Chloro-4-methylbenzene (Major) + 1-Chloro-2-methylbenzene (Minor)	
Conditions: Anhydrous AlCl3 as Lewis acid	
Product/Note: Para product major; halogen directs EAS to ortho and para	

9.3 Reactions with Metals — Wurtz-Fittig & Fittig

Wurtz-Fittig Reaction	Type: Organometallic Coupling
Ar-X + R-X + 2Na --[dry ether]--> Ar-R + 2NaX	
Conditions: Mixture of aryl and alkyl halide; Na in dry ether	
Product/Note: Gives alkylarene (Ar-R); useful for making arylalkanes	

Fittig Reaction	Type: Organometallic Coupling
2 Ar-X + 2Na --[dry ether]--> Ar-Ar + 2NaX	
Conditions: Only aryl halides; Na in dry ether	

Product/Note: Gives biphenyl type compounds (two aryl groups joined)

10. POLYHALOGEN COMPOUNDS

Compound	Formula	Common Name	IUPAC Name	Key Uses	NEET Notes
Dichloromethane	CH2Cl2	Methylene chloride	Dichloromethane	Solvent, paint remover, aerosol propellant, drug manufacturing	Harms CNS; exposure → dizziness, numbness
Chloroform	CHCl3	Chloroform	Trichloromethane	Solvent for fats/alkaloids/iodine; manufacture of Freon R-22; former anaesthetic	Oxidised by air+light → PHOSGENE (COCl2 — poisonous); stored in DARK bottles filled completely
Iodoform	CHI3	Iodoform	Triiodomethane	Former antiseptic (antiseptic due to liberation of FREE IODINE — not iodoform itself)	Objectionable smell; replaced by iodine formulations; yellow solid; iodoform test for methyl ketones
Carbon tetrachloride	CCl4	Carbon tetrachloride	Tetrachloromethane	Refrigerants, aerosol propellants, cleaning fluid, fire extinguisher	Causes liver cancer; depletes OZONE LAYER; rises to atmosphere and initiates ozone-depleting chain reactions
Freons	CCl2F2 (Freon-12)	Freons	Chlorofluorocarbons (CFCs)	Aerosol propellants, refrigeration, air conditioning	Made by Swarts reaction; extremely stable; depletes ozone layer in stratosphere via radical chain reactions
DDT	C14H9Cl5	DDT	p,p'-Dichlorodiphenyltrichloroethane	Insecticide (malaria mosquito, typhus lice)	Paul Muller — Nobel Prize 1948; fat-soluble, bioaccumulates; banned in USA 1973; insect resistance developed

📌 Chloroform Storage — NEET Favourite
Chloroform oxidised by O2 in presence of light → Phosgene (COCl2 = Carbonyl chloride — extremely poisonous)
Chemical: 2CHCl3 + O2 --[light]--> 2COCl2 + 2HCl
Storage: dark coloured bottles, completely filled (no air space)
Phosgene was used as chemical weapon in World War I
Ethanol added as preservative to decompose any phosgene formed

📌 Ozone Depletion — CCl4 & Freons
CCl4 released into air → rises to stratosphere → depletes ozone
Freons (CFCs) — extremely stable, reach stratosphere unchanged
In stratosphere: Freons initiate RADICAL CHAIN REACTIONS → upset ozone balance
Ozone depletion → increased UV exposure → skin cancer, eye diseases, immune disruption
Montreal Protocol: phased out use of CFCs worldwide

11. COMPLETE REACTION SUMMARY TABLE

Reaction Name	Reactants	Product	Reaction Type	Key Condition
R-OH + HCl	ROH + HCl	R-Cl	Nucleophilic Sub.	ZnCl2 (for 1°, 2°); none for 3°
R-OH + SOCl2	ROH + SOCl2	R-Cl (pure)	Nuc. Sub.	Best method; SO2+HCl escape
Finkelstein	R-Cl/Br + NaI	R-I + NaCl↓	Halogen Exchange (SN2)	Dry acetone; NaCl precipitates
Swarts	R-Cl + AgF/Hg2F2	R-F	Halogen Exchange	Metallic fluoride
Alkene + HX (Markovnikov)	CH3CH=CH2 + HI	CH3CHICH3 (major)	Electrophilic Addition	Normal addition; Markovnikov's rule
Alkene + HBr (Anti-Markovnikov)	CH3CH=CH2 + HBr + peroxide	CH3CH2CH2Br (major)	Free Radical Addition	Peroxide/hv; only HBr shows this
Alkene + Br2	CH2=CH2 + Br2/CCl4	BrCH2CH2Br	Electrophilic Addition	vic-Dibromide; test for unsaturation
Free Radical Halogenation	RH + X2	R-X (mixture)	Free Radical Sub.	UV light/heat; gives mixture
SN2 (NaOH)	CH3Cl + NaOH	CH3OH + Cl-	Nucleophilic Sub. (SN2)	Strong nucleophile; 2nd order; inversion
SN1 (tert-BuBr)	(CH3)3CBr + OH-	(CH3)3COH	Nucleophilic Sub. (SN1)	Polar protic; 1st order; racemisation
Beta-Elimination	R-CH2CH2X + alc. KOH	R-CH=CH2	Elimination (E2)	Alcoholic KOH + heat; Zaitsev's rule
Grignard Formation	R-X + Mg	RMgX	Organometallic	Dry ether; anhydrous
Wurtz	2RX + 2Na	R-R	Coupling (Organometallic)	Dry ether; doubles C atoms
Sandmeyer (Cl/Br)	ArNH2 → ArN2+ + CuX	Ar-X	Nuc. Sub. via diazonium	Cold 273-278K; CuCl or CuBr
Aryl iodide	ArN2+X- + KI	Ar-I	Nuc. Sub.	No Cu catalyst

				needed for I
Balz-Schiemann (F)	ArN2+BF4- → heat	Ar-F	Thermal decomp.	Only method for aryl fluoride
Halogenation of Arene	C6H6 + Cl2	C6H5Cl	EAS	Fe or FeCl3; dark; no F2 or I2 directly
Wurtz-Fittig	ArX + RX + 2Na	Ar-R	Coupling	Dry ether
Fittig	2ArX + 2Na	Ar-Ar	Coupling	Dry ether; biphenyl-type
Chlorobenzene hydrolysis	C6H5Cl + NaOH(aq)	C6H5OH	Nuc. Aromatic Sub.	623K, 300 atm — harsh conditions

12. NEET EXAM FOCUSED TRICKS & MCQ POINTS

📌 BOILING POINT TRENDS — Quick Revision
Trend 1: Same alkyl group — RI > RBr > RCl > RF (larger halogen = higher b.p.)
Trend 2: Same halogen, longer chain = higher b.p.
Trend 3: Branching DECREASES b.p. (n-BuBr > sec-BuBr > tert-BuBr)
Trend 4: Dihalobenzene isomers — nearly same b.p., but para has HIGHEST m.p.
Exception: Fluorides have LOW b.p. despite high polarity (weak van der Waals due to small F)

📌 REACTIVITY ORDERS — Must Memorise
SN2: CH3X > 1° > 2° > 3° [primary most reactive; methyl = fastest]
SN1: 3° > 2° > 1° > CH3X [tertiary most reactive; carbocation stability]
Halide leaving ability: I- > Br- > Cl- > F- (I = best leaving group)
R-X bond strength: R-F > R-Cl > R-Br > R-I (RF hardest to break)
R-X reactivity: R-I > R-Br > R-Cl >> R-F (RI most reactive)
Alcohol reactivity with HX: 3° > 2° > 1° (Lucas test order)
Grignard base strength: all equally reactive with acidic H sources
Allylic/Benzylic: HIGH SN1 (resonance-stabilised carbocation)
Vinylic/Aryl: LOWEST nucleophilic substitution reactivity

📌 NAMED REACTIONS — Full List for NEET
1. Finkelstein: RX + NaI (dry acetone) → RI + NaX↓ [iodide]
2. Swarts: RX + AgF/Hg2F2/CoF2/SbF3 → RF [fluoride]
3. Wurtz: 2RX + 2Na (dry ether) → RR + 2NaX [doubles C]
4. Grignard: RX + Mg (dry ether) → RMgX [organometallic]
5. Sandmeyer: ArNH2 → ArN2+ + CuX → ArX [Cl or Br]

6. Balz-Schiemann: $ArN_2^+BF_4^-$ → ArF [fluoride on ring]
7. Wurtz-Fittig: ArX + RX + 2Na → Ar-R [arylalkane]
8. Fittig: 2ArX + 2Na → Ar-Ar [biphenyl]
9. Markovnikov's Rule: HX addition to alkene — major product
10. Kharash/Anti-Markovnikov: HBr + peroxide → anti-Markovnikov (ONLY HBr)
11. Zaitsev's Rule: Dehydrohalogenation — more substituted alkene = major
12. Walden Inversion: stereochemical inversion in SN2

📌 IMPORTANT CONDITIONS TO REMEMBER
$SOCl_2$ → best reagent for ROH to RCl (gives pure product)
KOH (aqueous) → substitution (gives alcohol)
KOH (alcoholic) + heat → elimination (gives alkene)
HBr + peroxide → anti-Markovnikov; HBr alone → Markovnikov
HI: peroxide has NO effect (anti-Markovnikov not observed)
Br_2/CCl_4 → electrophilic addition to alkene (test for C=C)
Cl_2/UV light → free radical substitution on alkane
$Cl_2/FeCl_3$ (dark) → electrophilic substitution on benzene ring
Grignard: MUST use dry ether (even trace water destroys it)
Finkelstein: dry acetone (NaX is insoluble, precipitates, drives equilibrium)
SN1: polar protic solvent; SN2: polar aprotic solvent preferred
Sandmeyer: cold (273-278K) for diazonium salt stability
Chloroform: dark bottles, completely filled (prevents phosgene formation)

📌 COMMON NEET MISTAKES TO AVOID
DDT was discovered by Paul Muller (NOT prepared by him — prepared in 1873; discovered as insecticide in 1939)
Iodoform's antiseptic property is due to LIBERATED IODINE, not iodoform itself
C-Cl has HIGHEST dipole moment (NOT C-F) among methyl halides
H_2SO_4 NOT used with KI — it oxidises I^- to I_2 (use H_3PO_4 instead)
Peroxide effect: ONLY with HBr, NOT HCl or HI
$-NO_2$ at META position: NO effect on reactivity of aryl halide
p-Dichlorobenzene: highest MELTING POINT (not boiling point) among isomers
Propyl chloride ($n-C_3H_7Cl$) is LIGHTER than water (density 0.89 g/mL)
Chloroform stored in dark — oxidation to phosgene, not light-induced decomposition
SN2: inversion; SN1: racemisation (NOT inversion)
KCN gives nitrile (via C); AgCN gives isocyanide (via N)

📌 MCQ-ORIENTED FACTS
Thionyl chloride (SOCl2) — best to convert alcohol to alkyl chloride
Grignard reagent discovered by Victor Grignard (1900); Nobel Prize 1912
DDT: first chlorinated organic insecticide; banned in USA 1973
Paul Muller: Nobel Prize in Medicine and Physiology 1948 (for DDT as insecticide)
Freon 12 = CCl2F2; made by Swarts reaction from CCl4
William Nicol (1768-1851): developed first prism for plane polarised light
Van't Hoff: first Nobel Prize in Chemistry 1901 (solutions work)
Hughes worked under Ingold (University of London)
Halogenated compounds persist in environment — resist breakdown by soil bacteria
Chlorambucil: alkylating agent in cancer chemotherapy (contains Cl)
Halothane: anaesthetic — CF3-CHBrCl
Zaitsev formulated his rule in 1875

13. QUICK REVISION FLOWCHART

STARTING FROM	PRODUCT	REAGENT/METHOD	REACTION TYPE
Alcohol (R-OH)	R-Cl (pure)	SOCl2	Nucleophilic Sub.
Alcohol (R-OH)	R-Cl	HCl + ZnCl2	Nucleophilic Sub.
Alcohol (R-OH)	R-Br	48% HBr, heat	Nucleophilic Sub.
Alcohol (R-OH)	R-I	NaI + H3PO4(95%)	Nucleophilic Sub.
Alkyl Cl/Br	R-I	NaI, dry acetone (Finkelstein)	Halogen Exchange
Alkyl Cl/Br	R-F	AgF/Hg2F2 (Swarts)	Halogen Exchange
Alkene	Haloalkane (Markov.)	HX	Electrophilic Addition
Alkene	1°-Haloalkane (Anti-M)	HBr + peroxide	Free Radical Addition
Alkene	vic-Dihalide	X2/CCl4	Electrophilic Addition
Alkane	Haloalkane (mixture)	X2, UV/heat	Free Radical Sub.
Haloalkane (R-X)	Alcohol	NaOH(aq)	SN2/SN1
Haloalkane (R-X)	Alkene	Alc. KOH, heat	Beta-Elimination
Haloalkane (R-X)	Nitrile (RCN)	KCN	SN2
Haloalkane (R-X)	Isonitrile (RNC)	AgCN	SN2
Haloalkane (R-X)	Ether	NaOR' (Williamson)	SN2
Haloalkane (R-X)	Grignard (RMgX)	Mg, dry ether	Organometallic
2 Haloalkanes	Higher alkane (R-R)	Na, dry ether (Wurtz)	Coupling

ArNH2 (aniline)	Ar-Cl or Ar-Br	NaNO2+HCl; then CuX (Sandmeyer)	Nuc. Sub.
ArNH2	Ar-I	NaNO2+HX; then KI	Nuc. Sub.
ArNH2	Ar-F	NaNO2+HBF4; heat (Balz-Schiemann)	Thermal Decomp.
Benzene/Arene	Halobenzene	X2 + Fe/FeCl3 (dark)	EAS
Halobenzene	o+p products	EAS reagents	EAS (o,p-director)
Halobenzene	Phenol	NaOH(aq), 623K, 300 atm	Nuc. Aromatic Sub.